openEHR EHRCON26

Modelling the Episode-Oriented Medical Record according to Solon in openEHR

Patterns for Longitudinal, Problem- and Episode-Centred Care

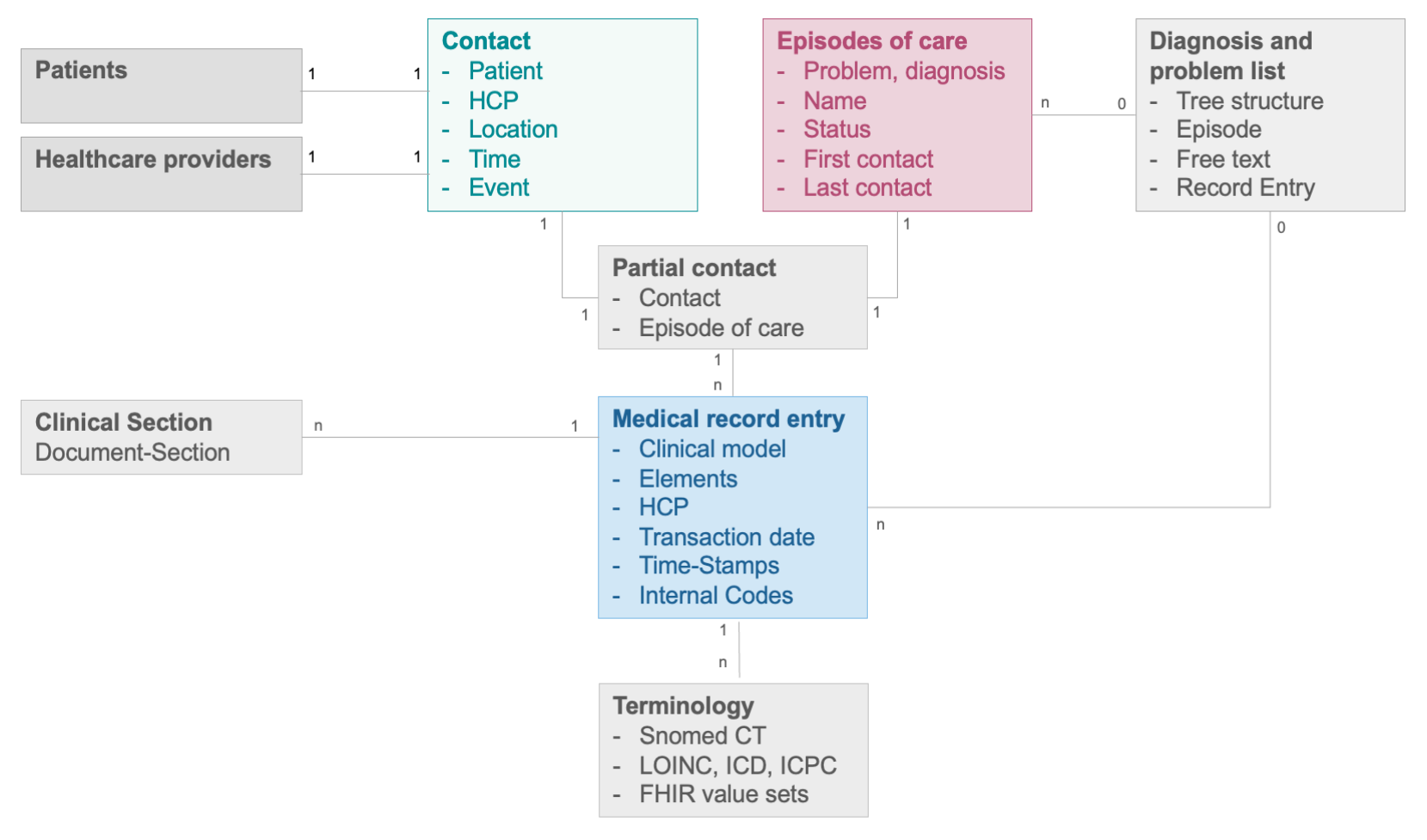
How can we organise an EHR around the longitudinal course of a health problem – rather than around documents and encounters?

From Solon's episode-oriented medical record to a computable openEHR architecture

Jean-Pierre Messerli · Independent Medical Informatics Researcher, Switzerland

The episode-oriented medical record

Three complementary contexts for longitudinal clinical documentation



Care contact

When was it documented?

Episode of Care

Which health problem?

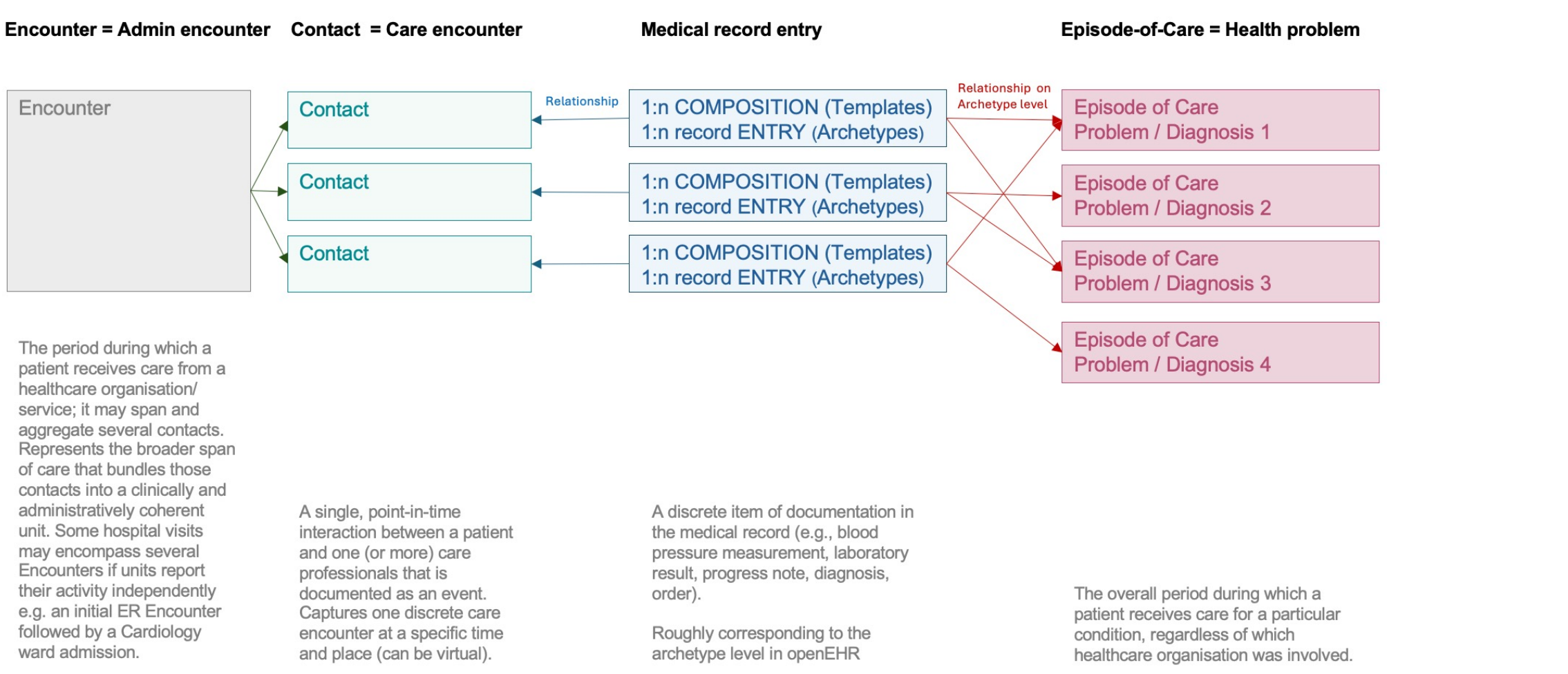
Clinical section

What type of information?

Every clinical entry is anchored in its event context, longitudinal health-problem context and clinical content context.

How can this be represented in openEHR?

COMPOSITIONs provide the building blocks – relationships provide the longitudinal structure



Episode COMPOSITIONs = longitudinal source of truth

Three implementation patterns

Three approaches to longitudinal relationships

FOLDER

Navigation

Intuitive patient-wide organisation

Governance of lifecycle, naming and updates

LINK

Referencing

Flexible explicit relationships

Higher query and application complexity

CLUSTER

Semantics

Strong semantic governance

Coded, queryable and reusable

EVALUATED WITH

132 clinical entries

8 clinician-centred user stories

Structured evaluation framework

What does this mean in practice?

A hybrid architecture enabling longitudinal clinical views without data duplication

RECOMMENDED HYBRID

Episode COMPOSITIONs
longitudinal source of truth

CLUSTERs
authoritative semantic representation

FOLDERs
patient-wide navigation

LINKs
selective explicit relationships

CLINICAL RELEVANCE

- Follow the course of a health problem over time
- Navigate between chronological, episode-centred and problem-list views
- Maintain consistent information without duplicating data

KEY CONTRIBUTION

Episode, diagnosis, problem and past medical history lists are derived clinical views rather than independently maintained data structures.

Episode-oriented care can be realised in openEHR without changing the Reference Model.

Thank you

Further resources, acknowledgements and contact

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VIDEO TUTORIALS

Session 1
High-Level EHR Organisation
https://vimeo.com/1167309403

Session 2
Modelling in openEHR
https://vimeo.com/1168024722

MASTERCLASS THESIS & ARTEFACTS

Data Model of the Episode-Oriented Medical Record according to Solon in openEHR

https://github.com/JPM-Consulting/openEHR-Masterclass-thesis
Thesis · Sample data · Supporting artefacts

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